

Ontology Standards and Semantic Integration in Industrial Model-Based Systems Engineering:
Mapping, Interoperability, and Empirical Evidence

Quick Reference

Key Findings Table

Aspect	Key Ontologies/Standards	Mapping to SysML/SysML v2	Empirical Evidence	Limitations/Challenges
Ontology Standards	PLIB, OWL, ISA-95, IEC 62264-2, TOGAF, Asset Administration Shell (AAS)	Model-driven engineering (EMF, QVT), semantic adaptation, OWL-based synchronization	Improved semantic interoperability, lifecycle management, decision support in aviation, manufacturing	Tool interoperability, inconsistent traceability, semantic grounding, organizational barriers
Upper Ontologies	BFO, SUO, PHYSSYS, DOLCE, EMMO	Foundation for domain ontologies, cross-layer mapping (e.g., SysDO)	Empirical validation in aircraft manufacturing, cross-domain integration	Limited standardization, need for tailored methodologies
Reference Ontologies	ISO/IEC/IEEE 15288-based, System Design Ontology (SysDO), GOPPRRE	Process-centric semantic frameworks, formalization of cross-layer relationships	Enhanced traceability, tool-agnostic interoperability	Gaps in quantitative evaluation, integration with legacy tools
Controlled Vocabularies	UNSPSC, eCl@ss, ISO/IEC 81346, MIL-STD-6016B	Transformed into OWL for MBSE, support for automated reasoning	Used in healthcare, manufacturing, defense for semantic integration	Transformation complexity, limited reuse across domains
Human-in-the-Loop	Cognitive Thread Ontology, collaborative ontology engineering	Formalizes stakeholder/process info, supports traceability	Improved traceability, reuse, collaborative design	Tool complexity, skill gaps, cultural resistance

Industrial Adoption	Aviation, manufacturing, healthcare, digital railways	Integration with SysML/SysML v2, digital twins, PLM	Lifecycle management, decision support, reduced errors	Immature toolchains, lack of standardized metrics
----------------------------	---	---	--	---

Direct Answer

Several ontology standards and frameworks are utilized in industrial MBSE and digital engineering to map heterogeneous information to system model elements, especially in SysML and SysML v2. Key approaches include domain-specific ontologies (PLIB, modular ISA-95), upper ontologies (BFO, SUO), reference ontologies (ISO/IEC/IEEE 15288-based), and controlled vocabularies (UNSPSC, eCl@ss) often transformed into OWL. Integration with SysML/SysML v2 is achieved via model-driven engineering (EMF, QVT) and semantic adaptation elements to ensure unambiguous interpretation and traceability. Empirical evidence, especially from aerospace and manufacturing, demonstrates improvements in lifecycle management, decision support, and interoperability; however, challenges persist in tool interoperability, standardized evaluation metrics, and organizational acceptance.

Study Scope

- **Time Period:** 2018–2024
- **Disciplines:** Systems engineering, manufacturing, aerospace, digital engineering, software engineering, healthcare, railways
- **Methods:** Systematic literature review, industrial case studies, empirical validation, tool and methodology benchmarking, qualitative and quantitative analysis

Assumptions & Limitations

- Most empirical evidence is qualitative or based on case studies; standardized, quantitative evaluation frameworks are rare.
- Industrial adoption is concentrated in large organizations (e.g., aerospace, manufacturing); SMEs face additional barriers.
- Tool interoperability and traceability mechanisms are still maturing; integration with legacy systems remains challenging.
- Cultural and organizational factors significantly affect adoption and effectiveness.

Suggested Further Research

- Develop standardized, quantitative metrics and evaluation frameworks for ontology-based MBSE.

- Advance ontology versioning, modularization, and integration with AI/ML for automated mapping.
- Bridge software engineering and other engineering disciplines through unified ontology frameworks.
- Expand training and knowledge transfer strategies to address skill gaps and cultural resistance.

1. Introduction

Semantic integration is a cornerstone of industrial Model-Based Systems Engineering (MBSE) and digital engineering, enabling the mapping of heterogeneous engineering information to system model elements—particularly in SysML and SysML v2. The proliferation of domain-specific tools, legacy data, and multidisciplinary teams necessitates robust ontology standards, upper ontologies, reference ontologies, and controlled vocabularies to ensure interoperability, traceability, and lifecycle management. Recent empirical studies highlight both the promise and persistent challenges of ontology-driven MBSE, especially regarding tool interoperability, traceability, and organizational adoption in complex industrial environments [1](#) [2](#) [3](#).

Scope and Motivation

This report synthesizes the state of the art in ontology standards and their application to mapping heterogeneous engineering information to SysML/SysML v2, with a focus on empirical evidence for interoperability, traceability, governance, human-in-the-loop applications, and industrial adoption. The analysis bridges theoretical frameworks with practical insights, drawing on recent industrial case studies and meta-analytical findings [1](#) [2](#) [3](#).

2. Ontology Standards, Upper Ontologies, and Controlled Vocabularies in Industrial MBSE

Key Ontology Standards and Languages

- **PLIB:** A specialized ontology language for engineering, supporting correctness and completeness in controlled environments. Managed via OntoDB/OntoQL, PLIB is widely used for product and resource information modeling [1](#).
- **ISA-95 & IEC 62264-2:** Modular ontologies based on these standards describe manufacturing systems, supporting runtime updates, semantic rule-based inference, and integration with Asset Administration Shells (AAS) for digital twins [4](#) [5](#) [6](#).
- **OWL (Web Ontology Language):** The de facto standard for ontology modeling, enabling semantic reasoning, interoperability, and transformation of controlled vocabularies into machine-readable formats [7](#) [8](#).
- **TOGAF:** Enterprise architecture metamodels formalized as ontologies to reduce ambiguity and facilitate knowledge discovery and interoperability [9](#).
- **Asset Administration Shell (AAS):** Provides a standardized digital twin framework, extended with ontologies for semantic interoperability and real-time data integration [6](#) [10](#).

Platforms and Toolkits

- **Protégé, EMF, QVT:** Widely used for ontology development, model transformation, and integration with MBSE tools [11](#) [12](#).
- **IEOP:** Aligns ontologies to industrial data sources and upper ontologies for enterprise data governance [13](#).

Upper and Reference Ontologies

- **BFO (Basic Formal Ontology):** Provides foundational categories for integrating domain ontologies, supporting cross-domain semantic interoperability [14](#) [15](#).
- **SUO (Standard Upper Ontology):** Used as a foundational framework for systems engineering ontologies [15](#).
- **PHYSSYS:** A formal upper ontology for dynamic physical systems, supporting reusable model components across disciplines [16](#).
- **DOLCE, EMMO:** Support integration of domain-specific ontologies, especially in computational engineering and materials science [17](#).
- **System Design Ontology (SysDO):** Extends BFO/IOF-core, formalizing cross-layer relationships in aircraft manufacturing and enabling tool-agnostic interoperability [2](#).

Controlled Vocabularies and Taxonomies

- **UNSPSC, eCI@ss, RosettaNet:** Serve as inputs for domain ontologies, often transformed into OWL for semantic integration in MBSE [8](#).
- **ISO/IEC 81346:** Used as a structural foundation, requiring transformation into expressive OWL ontologies for automated reasoning [18](#).
- **MIL-STD-6016B:** Transformed into OWL for expert-system reasoning in defense MBSE applications [19](#).
- **Healthcare Standards (FHIR, HL7, SNOMED, LOINC, ICD-11):** Embedded in OWL ontologies for complex system-of-systems domains [20](#).

Domain-Specific and Modular Ontologies

- **Manufacturing:** Modular ISA-95 ontologies, ORMA ontology for asset/process/product knowledge, and AAS-based ontologies for real-time adaptive control [4](#) [10](#).
- **Aviation:** Ontology-based MBSE frameworks for lifecycle management and decision support [3](#).

- **Healthcare, Digital Railways:** Domain-specific ontologies for regulatory compliance and semantic integration [20](#) [21](#).
- **Systemic Ontology Language (SOL):** A proposed formal ontology modeling language for regulated industries [18](#).

Synthesis

Industrial MBSE leverages a rich ecosystem of ontology standards, upper ontologies, and controlled vocabularies, often transformed into OWL for semantic integration. These frameworks underpin the mapping and synchronization of heterogeneous engineering information, supporting lifecycle management, traceability, and cross-domain interoperability [1](#) [14](#) [15](#) [22](#).

3. Mapping Ontologies to SysML and SysML v2

Ontology-Driven Mapping and Synchronization

- **OWL Ontologies:** Enable automatic synchronization between heterogeneous domain models and SysML/SysML v2, reducing manual effort and data inconsistency [22](#) [23](#).
- **Semantic Adaptation Elements:** Address ambiguities in SysML v1, ensuring unambiguous interpretation and correct integration of models across tools and simulation platforms [24](#).
- **Formal Semantics:** Embedding SysML within formal logic and defining concrete semantics for SysML elements supports tool interoperability and verification [24](#) [25](#).

Model Transformation Techniques

- **Model-Driven Engineering (MDE):** Techniques such as EMF and QVT automate mapping from SysML v2 to standards like AAS, preserving structural and behavioral semantics [12](#).
- **XMI-Based Transformation:** Facilitates conversion of heterogeneous SysML data from various tools into standardized SysML objects [26](#).
- **LLM-Assisted Transformation:** Large language models show promise for automating model transformations but currently lack semantic richness and reliability for complex mappings [27](#).

Semantic Adaptation and Formal Semantics

- **Formalization of SysML Blocks:** Using process algebra (CML) and Petri nets to provide mechanized semantics and support automatic generation of formal models [25](#) [28](#).
- **Translation Rules:** Refinement of translation rules for SysML diagrams into formal languages (e.g., CSP#, Petri nets) enhances verification and tool interoperability [29](#) [30](#).

Domain-Specific Language Extensions

- **DSLs and SysML Profiles:** Extend SysML to capture specialized engineering knowledge, improving modularity, maintainability, and clarity in complex system designs [31](#) [32](#) [33](#).
- **Graphical Editors and APIs:** Enable domain-specific modeling and integration with CAD and other tools, supporting bi-directional synchronization [34](#) [35](#).

Synthesis

Ontology-driven frameworks, formal semantics, and model transformation techniques are central to mapping heterogeneous engineering information to SysML/SysML v2. These approaches enhance traceability, reduce ambiguity, and support automated synchronization, though challenges remain in semantic richness and tool interoperability [12](#) [22](#) [24](#).

4. Empirical Evidence: Interoperability, Traceability, Governance, and Industrial Adoption

Interoperability and Semantic Integration

- **Digital Thread and Digital Twin:** Ontology-driven approaches provide a semantic foundation for integrating heterogeneous models, supporting intra- and cross-organizational interoperability and trustworthy decision-making [36](#) [37](#).
- **Semantic Web Technologies:** Use of OWL and Description Logic enhances reasoning, reduces integration time and errors, and supports scalable digital twin implementations [37](#) [38](#).

Traceability and Lifecycle Management

- **Cognitive Thread Ontology:** Formalizes stakeholder, model, and process information, improving traceability and reuse in complex engineered systems [39](#).
- **Lifecycle Management:** Ontology-based MBSE frameworks improve lifecycle integration, decision support, and traceability, especially in aviation and aerospace [3](#) [40](#) [41](#).

Governance and Data Management

- **Ontology-Based Governance Models:** Enhance product lifecycle management, data reusability, and unified evolution in digital twin environments [42](#) [43](#).
- **Enterprise Architecture:** Ontologies formalize enterprise architecture processes, supporting process modeling, quality assessment, and program management [42](#).

Human-in-the-Loop and Collaborative Applications

- **Collaborative Ontology Engineering:** Human involvement improves accuracy and completeness of ontologies, supporting collaborative system design [44](#).
- **Human Factors Integration:** Ontologies facilitate communication between systems engineering and

human factors engineering, enhancing stakeholder engagement [45](#) [46](#).

Industrial Adoption and Limitations

- **Adoption in Aviation and Manufacturing:** Demonstrated improvements in lifecycle management, decision support, and reduced errors; however, tool maturity, traceability, and semantic grounding remain challenges [47](#) [48](#).
- **Barriers:** Organizational resistance, skill gaps, and lack of standardized metrics limit broader adoption [48](#) [49](#).

Synthesis

Empirical evidence underscores the benefits of ontology-based MBSE for interoperability, traceability, and governance, particularly in complex industries. However, persistent challenges in tool interoperability, traceability mechanisms, and organizational acceptance highlight the need for further research and standardization [36](#) [39](#) [42](#) [47](#).

5. Ontology Support for Integration and Mapping Challenges in MBSE

Shared Vocabularies and Automated Data Exchange

- **OWL Ontologies:** Standardize data management and enable automatic synchronization between SysML and heterogeneous domain models, reducing manual errors [22](#).
- **Semantic Web Technologies (OML):** Create semantic layers for uniform access, classification, and reuse of federated MBSE data [50](#).

Formalization of Cross-Layer Relationships

- **SysDO:** Formalizes cross-layer relationships in aircraft manufacturing, bridging semantic discontinuities and enabling tool-agnostic interoperability [2](#).
- **GOPPRRE Metamodel:** Supports unified model representations and ontological data integration across MBSE formalisms and lifecycle stages [51](#).

Interdisciplinary Communication and Integration

- **Shared Ontologies:** Enhance communication between systems engineering and human factors engineering by establishing common vocabularies and linking domain-specific terms [52](#).
- **Foundational Ontologies:** Promote interoperability and standardization across disciplines [53](#).

Semantic Interoperability in Simulation and Digital Twins

- **Ontology-Based Co-Simulation:** Combines SysML v2 with ontology-based semantics and graph databases for unified simulation platforms [54](#) [55](#).

- **Digital Twin Integration:** Ontologies enable semantic interoperability and reasoning in digital twin implementations, supporting automated maintenance and optimization [6](#).

Semantic Web Technologies and Ontology Modeling Languages

- **OWL and OML:** Facilitate semantic integration, traceability, and lifecycle management in MBSE toolchains, supporting automated reasoning and verification [50](#) [56](#) [57](#).

Synthesis

Ontologies address integration and mapping challenges in MBSE by providing shared vocabularies, formalizing cross-layer relationships, and enabling automated data exchange and reasoning. Semantic web technologies and ontology modeling languages are critical enablers for traceability and lifecycle management [2](#) [22](#) [56](#).

6. Gaps, Challenges, and Future Research Directions

Tool Interoperability and Traceability Barriers

- **Immature Tooling:** Limited interoperability and inconsistent traceability mechanisms hinder seamless integration and reuse [39](#) [58](#).
- **Semantic Grounding:** Insufficient semantic frameworks and verification mechanisms limit effective integration and traceability [59](#).

Cultural and Organizational Challenges

- **Cultural Resistance:** Organizational culture, resistance to change, and lack of skilled personnel impede adoption [60](#) [61](#) [62](#).
- **Stakeholder Engagement:** Complexity and tooling barriers restrict broader stakeholder involvement [63](#).

Metrics and Evaluation Frameworks

- **Lack of Standardization:** Few quantitative metrics or standardized evaluation frameworks exist for assessing ontology-based MBSE methodologies [64](#) [65](#).
- **Proposed Approaches:** Ontology-based architectural description frameworks, risk classification models, and resilience analysis methodologies are emerging [65](#) [66](#).

Bridging Software and Other Engineering Disciplines

- **Unified Ontology Frameworks:** Needed to bridge software engineering with other disciplines, supporting comprehensive lifecycle management [3](#) [67](#).
- **Modular Patterns:** Ontology-based patterns and modular blocks facilitate communication and integration

across domains [68](#).

Training and Knowledge Transfer Strategies

- **Scenario-Based Learning:** Effective MBSE training combines theory with practical scenarios, enhancing knowledge transfer [69](#).
- **Alignment with Industry Needs:** Academic curricula should reflect industry requirements, tools, and concepts [49](#).
- **Interdisciplinary Education:** Project-based and interdisciplinary learning frameworks support digital transformation readiness [70](#).

Synthesis

Persistent gaps in tool interoperability, traceability, and organizational culture limit the full realization of ontology-based MBSE. Addressing these challenges requires standardized metrics, unified frameworks, and comprehensive training strategies tailored to industrial contexts [3](#) [58](#) [60](#) [64](#) [69](#).

7. Conclusion

Summary of Insights and Recommendations

- **Diverse Ontology Ecosystem:** Industrial MBSE employs a range of ontology standards, upper ontologies, and controlled vocabularies, often transformed into OWL for semantic integration and mapping to SysML/SysML v2 [2](#).
- **Enhanced Interoperability and Traceability:** Ontology-driven frameworks, formal semantics, and model transformation techniques improve interoperability, traceability, and lifecycle management, with empirical validation in aviation, manufacturing, and other sectors [36](#).
- **Persistent Challenges:** Tool interoperability, traceability mechanisms, and organizational culture remain significant barriers to broader adoption and effectiveness [3](#).
- **Future Directions:** Standardized, quantitative evaluation frameworks, advanced ontology versioning, interdisciplinary integration, and comprehensive training are critical for advancing ontology-based MBSE in industry.

Actionable Recommendations:

- Invest in tool interoperability and semantic frameworks to support traceability and lifecycle integration.
- Develop and adopt standardized metrics and evaluation frameworks for ontology-based MBSE.
- Foster organizational culture change and provide targeted training to bridge skill gaps.
- Pursue interdisciplinary research and unified ontology frameworks to bridge software and other

engineering disciplines.

In summary, ontology standards and semantic integration are foundational to the future of industrial MBSE and digital engineering. Continued research, standardization, and organizational commitment are essential to realize their full potential for mapping, interoperability, and lifecycle management in complex engineering environments.

References

1. Ontologies in engineering: the OntoDB/OntoQL platform Ait-Ameur, Y., Baron, M., Bellatreche, L., (...), Sardet, E. Soft Computing, 2017 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/84924561280?origin=scopusAI>
2. SysDO: A Domain-Level Ontology for Cross-Domain System Design in Aircraft Manufacturing Liu, Z., Namaki-Araghi, S., Sarkar, A., (...), Karray, M.H. CEUR Workshop Proceedings, 2025 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105039590918?origin=scopusAI>
3. Insights into ontology-based model-based systems engineering: state of the art and enabling framework Dong, M., Wang, G., Lu, J., (...), Kiritsis, D. Advanced Engineering Informatics, 2026 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105041222744?origin=scopusAI>
4. An ISA-95 based Ontology for Manufacturing Systems Knowledge Description Extended with Semantic Rules Seyedamir, A., Ferrer, B.R., Lastra, J.L.M. Proceedings - IEEE 16th International Conference on Industrial Informatics, INDIN 2018, 2018 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85055527056?origin=scopusAI>
5. A graphical toolkit for IEC 62264-2 Lang, L., Wally, B., Huemer, C., (...), Wimmer, M. Procedia CIRP, 2020 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85092435541?origin=scopusAI>
6. Enabling semantic interoperability of asset administration shells through an ontology-based modeling method Huang, Y., Dhouib, S., Medinacelli, L.P., Malenfant, J. Proceedings - ACM/IEEE 25th International Conference on Model Driven Engineering Languages and Systems, MODELS 2022: Companion Proceedings, 2022 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85142934810?origin=scopusAI>
7. Semantics enhanced engineering and model reasoning for control application development Hästbacka, D., Kuikka, S. Multimedia Tools and Applications, 2013 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/84876131531?origin=scopusAI>
8. Products and Services Ontologies: A methodology for deriving OWL Ontologies from Industrial categorization standards Hepp, M. International Journal on Semantic Web and Information Systems, 2006 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/33746763436?origin=scopusAI>
9. On The Formalization of The TOGAF Content MetaModel Using Ontologies El Idrissi, B., Tetou, C., Doumi, K. 2022 International Conference on Intelligent Systems and Computer Vision, ISCV 2022, 2022 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85134300854?origin=scopusAI>
10. Leveraging Semantic Knowledge Graphs for Smart Manufacturing Integration via Asset Administration Shells Kheshfeh, M.W., Linke, S., Müller, M., (...), Mecke, R. Proceedings of International Conference on Applied Innovation in IT, 2025 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105040028343?origin=scopusAI>

11. Ontology-based solutions for interoperability among product lifecycle management systems: A systematic literature review Fraga, A.L., Vegetti, M., Leone, H.P. Journal of Industrial Information Integration, 2020 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85095454375?origin=scopusAI>
12. From engineering models to digital twins: Generating AAS from SysML v2 models Ferko, E., Berardinelli, L., Bucaioni, A., (...), Wimmer, M. Journal of Systems and Software, 2026 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105023655522?origin=scopusAI>
13. Building an Industrial Ontology Engineering Platform Brynildsen, M.H., Jakobsen, C., Abildgaard, N., Woods, C. CEUR Workshop Proceedings, 2023 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85184359422?origin=scopusAI>
14. Part 2: Ontology development, tools and languages Mizoguchi, R. New Generation Computing, 2004 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/0742268909?origin=scopusAI>
15. The Ontology of Systems Engineering: Towards a Computational Digital Engineering Semantic Framework Orellana, D., Mandrick, W. Procedia Computer Science, 2019 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85074978388?origin=scopusAI>
16. Engineering ontologies Borst, P., Akkermans, H., Top, J. International Journal of Human Computer Studies, 1997 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/0031078545?origin=scopusAI>
17. SEMANTIC INTEROPERABILITY BASED ON THE EUROPEAN MATERIALS AND MODELLING ONTOLOGY AND ITS ONTOLOGICAL PARADIGM: MEREOSEMIOTICS Horsch, M.T., Chiacchiera, S., Schembera, B., (...), Todorov, I.T. World Congress in Computational Mechanics and ECCOMAS Congress, 2021 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85102974749?origin=scopusAI>
18. A Co-Constructed Formal Ontology Modelling Language to Enhance Interoperability Barre, T., Sauvage, M., Philippe, V., Tonneau, M. ISSE 2025 - 11th IEEE International Symposium on Systems Engineering, Symposium Proceedings, 2025 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105034966107?origin=scopusAI>
19. GraphRAG-ASCOC: A lightweight framework for adaptive synonym-aware clustering and ontology completion Wang, D., Tyszberowicz, S., Han, P., (...), Liu, B. Expert Systems with Applications, 2026 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105029551112?origin=scopusAI>
20. An Ontology-Based Architecture for Interoperable Healthcare Systems-of-Systems: Structure, Interaction Patterns, and Covenant-Based Governance Mogahed, M., Mansouri, M. Systems, 2026 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105037058569?origin=scopusAI>
21. Formation of an Ontology-Based Model of Interaction Between the Railway Industry and Engineering Education Khabarov, V., Volegzhanina, I. Lecture Notes in Networks and Systems, 2022 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85127269071?origin=scopusAI>
22. Development of an OWL Ontology Based on the Function-Oriented System Architecture to Support Data Synchronization Between SysML and Domain Models Zhang, Y., Jacobs, G., Zhao, J., (...), Hoepfner, G. International Joint Conference on Knowledge Discovery, Knowledge Engineering and Knowledge Management, IC3K - Proceedings, 2023 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85179556199?origin=scopusAI>
23. An approach for holistic model-based engineering of industrial plants Hooshmand, Y., Adamenko, D., Kunnen, S., Köhler, P. Proceedings of the International Conference on Engineering Design, ICED, 2017

<https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85029806056?origin=scopusAI>

24. Multi-paradigm semantics for simulating SysML models using SystemC-AMS Cafe, D.C., Dos Santos, F.V., Hardebolle, C., (...), Boulanger, F. Forum on Specification and Design Languages, 2013 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/84891341408?origin=scopusAI>

25. Formal models of SysML blocks Miyazawa, A., Lima, L., Cavalcanti, A. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 2013 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/84889560439?origin=scopusAI>

26. XMI-Based SysML Model Transformation Li, Y., Zhang, H., Lv, J., Liu, Y. 2022 7th International Conference on Control, Robotics and Cybernetics, CRC 2022, 2022 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85149314404?origin=scopusAI>

27. Learning to Transform: Evaluating LLMs on Model Transformation by Example Dao, D., Bucaioni, A., Cicchetti, A. Proceedings - 2025 ACM/IEEE 28th International Conference on Model Driven Engineering Languages and Systems Companion, MODELS-C 2025, 2025 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105030486105?origin=scopusAI>

28. Recursive ECATNets-based approach for formally verifying system modelling language activity diagrams Rahim, M., Kheldoun, A., Boukala-Ioualalen, M., Hammad, A. IET Software, 2015 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/84943232486?origin=scopusAI>

29. Translation rules of SysML state machine diagrams into CSP# toward formal model checking Ando, T., Yatsu, H., Kong, W., (...), Fukuda, A. International Journal of Web Information Systems, 2014 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/84907143239?origin=scopusAI>

30. Towards the formal verification of SysML specifications: Translation of activity diagrams into modular Petri nets Rahim, M., Hammad, A., Boukala-Ioualalen, M. Proceedings - 3rd International Conference on Applied Computing and Information Technology and 2nd International Conference on Computational Science and Intelligence, ACIT-CSI 2015, 2015 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/84962627113?origin=scopusAI>

31. A method for customizing aerospace equipment views based on domain modeling language Sang, S., Xu, J., Yuan, W., (...), Liang, Z. Xi Tong Gong Cheng Yu Dian Zi Ji Shu/Systems Engineering and Electronics, 2026 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105039973199?origin=scopusAI>

32. Domain-specific languages and diagram customization for a concurrent engineering environment Cole, B., Dubos, G., Banazadeh, P., (...), Picha, F. IEEE Aerospace Conference Proceedings, 2013 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/84878729407?origin=scopusAI>

33. Industrial application of formal models generated from domain specific languages Hooman, J. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 2016 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/84962531698?origin=scopusAI>

34. Architecture to geometry - Integrating system models with mechanical design Bajaj, M., Zwemer, D., Cole, B. AIAA Space and Astronautics Forum and Exposition, SPACE 2016, 2016 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85088771602?origin=scopusAI>

35. Development and Implementation of Digital Heterogeneous Models in Model-based Systems Engineering Schumacher, T., Stephan, R., Inkermann, D. Procedia CIRP, 2025 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105015300842?origin=scopusAI>

36. Exploring Models' Interoperability in Digital Thread and Twin: A Proposal for an Ontology-Driven

Approach Based on Intra-And Cross-Organizational Enterprise Integration Liu, Z., Sarkar, A., Namaki Araghi, S., (...), Karray, M.H. Lecture Notes in Production Engineering, 2025 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105014218503?origin=scopusAI>

37. Semantic foundations for digital twins: the contribution of ontological analysis Elhajj, M. Frontiers in Computer Science, 2026 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105032771143?origin=scopusAI>

38. Ensuring the Connection Between Physical and Virtual Models in the Context of Digital Twins Abadi, M., Abadi, A., Abadi, C., Ben-Azza, H. Lecture Notes in Networks and Systems, 2023 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85161418561?origin=scopusAI>

39. Design ontology for cognitive thread supporting traceability management in model-based systems engineering Wu, S., Wang, G., Lu, J., (...), Kiritsis, D. Journal of Industrial Information Integration, 2024 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85192136305?origin=scopusAI>

40. Model-Based Systems Engineering Approach for a Systematic Design of Aircraft Engine Inlet Jagtap, S.S., Strehlow, A., Reitz, M., (...), Cinar, G. AIAA Science and Technology Forum and Exposition, AIAA SciTech Forum 2025, 2025 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105001312303?origin=scopusAI>

41. Using a model based systems engineering approach for aerospace system requirements management Subarna, Jawale, A.K., Vidap, A.S., (...), Myla, S. AIAA/IEEE Digital Avionics Systems Conference - Proceedings, 2020 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85097987442?origin=scopusAI>

42. Leveraging Taxonomies and Enterprise Architecture for Enhanced MBSE and Digital Engineering Practice Hugues, J., Shevchenko, N., Dempsey, C. AIAA Science and Technology Forum and Exposition, AIAA SciTech Forum 2026, 2026 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105031079873?origin=scopusAI>

43. Data Evolution Governance for Ontology-Based Digital Twin Product Lifecycle Management Ren, Z., Shi, J., Imran, M. IEEE Transactions on Industrial Informatics, 2023 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85133732200?origin=scopusAI>

44. From Human- to LLM-Centered Collaborative Ontology Engineering Doumanas, D., Bouchouras, G., Soularidis, A., (...), Vouros, G. Applied Ontology, 2025 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105036508065?origin=scopusAI>

45. Expanding mbse to incorporate human systems integration modelling Rountree, I., Thomas, L.D. AIAA Scitech 2021 Forum, 2021 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85100423118?origin=scopusAI>

46. Extending model-based system engineering for human machine interaction analysis and fault tolerant design Orellana, D.W., Madni, A.M. AIAA Infotech at Aerospace Conference and Exhibit 2012, 2012 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/84880784757?origin=scopusAI>

47. Combining model-based systems engineering, simulation and domain engineering in the development of industrial automation systems: Industrial case study Scheeren, I., Pereira, C.E. Proceedings - IEEE 17th International Symposium on Object/Component/Service-Oriented Real-Time Distributed Computing, ISORC 2014, 2014 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/84936861058?origin=scopusAI>

48. Industry-Driven Model-Based Systems Engineering (MBSE) Workforce Competencies—An AI-Based Competency Extraction Framework Akundi, A., Ravipati, P.R.T., Luna Fong, S.A., Otieno, W. Systems,

2025 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105017428550?origin=scopusAI>

49. Mapping industry workforce needs to academic curricula – A workforce development effort in model-based systems engineering Akundi, A., Ankobiah, W. Systems Engineering, 2024 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85184157973?origin=scopusAI>

50. Building a Semantic Layer for Early Design Trade Studies in the Development of Commercial Aircraft Zindel, A., Feo-Arenis, S., Helle, P., (...), Elaasar, M. ISSE 2022 - 2022 8th IEEE International Symposium on Systems Engineering, Conference Proceedings, 2022 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85146928127?origin=scopusAI>

51. Design Ontology Supporting Model-Based Systems Engineering Formalisms Lu, J., Ma, J., Zheng, X., (...), Kiritsis, D. IEEE Systems Journal, 2022 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85114742618?origin=scopusAI>

52. Importance of Ontologies for Systems Engineering (SE) and Human Factors Engineering (HFE) Integration Rudder, S., Herber, D.R. Applied Human Factors and Ergonomics International, 2024 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105031052281?origin=scopusAI>

53. Benefits of realist ontologies to systems engineering1 Merrell, E.C., Kelly, R.M., Kasmier, D., (...), Bush, K. CEUR Workshop Proceedings, 2021 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85117009766?origin=scopusAI>

54. Exploring ontology-based approach to facilitate integration of multi-physics and visualization for numerical models Vucinic, D., Pesut, M., Jovic, F., Lacor, C. Proceedings of the ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference 2009, DETC2009, 2010 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/77953718606?origin=scopusAI>

55. Bridging conceptual models and numerical domains: a semantic integration approach for federated co-simulation in MBSE Zimmermann, T.C., Cederbladh, J., Konietzko, E.P., Lünemann, P. Software and Systems Modeling, 2026 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105043945854?origin=scopusAI>

56. An approach for detecting and resolving inconsistency using DL rules for OWL generation from UML models Pabitha, P., Shobana Priya, A.M., Rajaram, M. European Journal of Scientific Research, 2012 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/84859258645?origin=scopusAI>

57. Sharing Ontologies and Rules Using Model Transformations Milanović, M., Djurić, D., Gašević, D., Devedžić, V. Handbook of Research on Emerging Rule-Based Languages and Technologies: Open Solutions and Approaches: Volumes 1-2, 2009 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105012654201?origin=scopusAI>

58. Should i stay or should i go?: On forces that drive and prevent mbse adoption in the embedded systems industry Vogelsang, A., Amorim, T., Pudlitz, F., (...), Philipps, J. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 2017 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85034595909?origin=scopusAI>

59. A Qualitative Analysis of MBSE Methodologies: Traceability and Early Design Phase Integration Pasquariello, A., Rega, A., Vitolo, F., Patalano, S. Lecture Notes in Mechanical Engineering, 2026 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105030247880?origin=scopusAI>

60. A survey on the utilization of ontologies for decision-making in industrial product development Pickel, J., Goetz, S., Wartzack, S. Procedia CIRP, 2025 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105015303444?origin=scopusAI>

61. Implementation and Assessment of a Comprehensive Model-Based Systems Engineering Methodology with Regard to User Acceptance in Practice Mandel, C., Kaspar, J., Heitmann, R., (...), Albers, A. *Procedia CIRP*, 2023 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85169887924?origin=scopusAI>
62. Model-based systems engineering implementation considerations Vaneman, W.K., Carlson, R. *SysCon 2019 - 13th Annual IEEE International Systems Conference, Proceedings*, 2019 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85073167601?origin=scopusAI>
63. Navigating viewpoints in MBSE: challenges, potential and pathways for stakeholder participation in industry Förster, F., Koldewey, C., Bernijazov, R., (...), Bursac, N. *Proceedings of the Design Society*, 2025 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105022777983?origin=scopusAI>
64. MBSE methodologies Estefan, J.A., Weilkiens, T. *Handbook of Model-Based Systems Engineering*, 2023 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85196235165?origin=scopusAI>
65. An Architectural Description for the Application of Mbse in Complex Systems Odukoya, K.A., Whitfield, R.I., Hay, L., (...), Robb, M. *ISSE 2021 - 7th IEEE International Symposium on Systems Engineering, Proceedings*, 2021 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85119098729?origin=scopusAI>
66. A hetero-functional graph resilience analysis for convergent systems-of-systems Farid, A.M. *International Journal of Systems Science*, 2025 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105025468902?origin=scopusAI>
67. Toward Demystifying the Missing Links in Model-Based Systems Engineering (MBSE) Khandoker, A., Sint, S., Gessl, G., Zeman, K. *Systems*, 2026 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105031063195?origin=scopusAI>
68. Ontology-Based Pattern for System Engineering Ernadote, D. *Proceedings - ACM/IEEE 20th International Conference on Model Driven Engineering Languages and Systems, MODELS 2017*, 2017 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/85040616641?origin=scopusAI>
69. Development of a Practical MBSE Training Demir, M., Höfermann, H. *ZWF Zeitschrift fuer Wirtschaftlichen Fabrikbetrieb*, 2025 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105023162084?origin=scopusAI>
70. Applying Model-Based Systems Engineering To Project Based Learning: A Framework For Systems Thinking & Design Integration Rathi, M., Winkle, C., Sharma, I., (...), Abdoli, S. *IEEE Global Engineering Education Conference, EDUCON*, 2026 <https://www-scopus-com.ezproxy.bib.hm.edu/pages/publications/105044409545?origin=scopusAI>